Question:

Toulouse has twice as many sheep as Charleston. Charleston has 4 times as many sheep as Seattle. How many sheep do Toulouse, Charleston, and Seattle have together if Seattle has 20 sheep?

Reasoning + Answer:

<think> To solve this problem, let's first define a variable to help us work through the relationships between the number of sheep in Toulouse, Charleston, and Seattle. Let's say \(S \) represents the number of sheep in Seattle. According to the problem, \(S = 20 \). Next, we use the information that Charleston has 4 times as many sheep as Seattle. So, we can express the number of sheep in Charleston (\(C \)) in terms of \(S \). According to the problem, \(C = 4 \times S \). Substituting the value of \(S \) (which is 20) into this equation, we get \(C = 4 \times 20 \). Let's calculate \(4 \times 20 \) to find the number of sheep in Charleston: \(4 \times 20 = 80 \). So, Charleston has 80 sheep. Now, let's focus on Toulouse. The problem states that Toulouse has twice as many sheep as Charleston. Let's denote the number of sheep in Toulouse as \(T \). According to the problem, \(T = 2 \times C \). We already found that \(C = 80 \), so we substitute 80 for \(C \) in the equation: \(T = 2 \times 80 \). Now, let's calculate \(2 \times 80 \) to find the number of sheep in Toulouse: \(2 \times 80 = 160 \). So, Toulouse has 160 sheep. Finally, to find out how many sheep Toulouse, Charleston, and Seattle have together, we need to add the number of sheep in all three cities. We have \(S = 20 \) for Seattle, \(C = 80 \) for Charleston, and \(T = 160 \) for Toulouse. Adding these together: \(20 + 80 + 160 \). Let's perform this addition: \(20 + 80 = 100 \), and then \(100 + 160 = 260 \). </think> <answer> 260 </answer>